

retsupp — meeting summary

for discussion with Dock Duncan & Jan Theeuwes

WHAT WE FOUND (model-free, robust to multiple analyses)

- HP-specific PRF push-away in V3AB and hV4, paired HP–Opposite metric (no Jensen):
V3AB $p_{\text{FDR}} = 0.016$ hV4 $p_{\text{FDR}} = 0.008$
Combined V3AB+hV4 a priori test: $t = 4.18$, $p = 0.0003$
- Direction asymmetry rules out Jensen's-inequality bias:
in V3AB and hV4 the OPPOSITE-anchored shift goes NEGATIVE while HP-anchored is positive.
Pure noise/Jensen would put both positive.
- Projection-vs-distance with kernel/LOWESS smoother + multi- R^2 thresholds:
V3AB: negative cluster at $d \approx 4\text{--}6^\circ$ (relative attraction at far distance)
 $p = 0.018, 0.020, 0.050, 0.043$ across $r^2 > \{0.10, 0.20, 0.30, 0.40\}$
V0: positive cluster at $d \approx 2\text{--}3^\circ$ (suppression at intermediate distance)
 $p = 0.003$ at $r^2 > 0.10$ and $r^2 > 0.30$
T0: trending negative cluster at $d \approx 4\text{--}5^\circ$ (matches V3AB direction)
hV4: strong near-HP suppression visible in paired metric ($p < 0.01$) but
localized at $d < 1^\circ$ so dispersed in coarser smoothings
- None of this depends on aperture filter or R^2 threshold – survives sensitivity sweep.

WHAT'S NEXT (cluster analyses, all submitted/staged today)

1. GLMsingle single-trial betas – replicating Richter 2025 BOLD-suppression pattern at HPDL using location-localizer ROIs.
SLURM array 2747605 (subjects 1-30, running NOW).
2. Extended-design PRF fits with distractors as additional stimulus inputs.
Tests whether the bar-aperture-edge signal is real or a fitting artefact.
SLURM array 2748981 (subjects 1-30, queued).
3. Per-ROI hierarchical Bayesian GAM (bambi/HSGP):
 $\text{proj} \sim s(\text{distance}) + (1 \mid \text{subject})$
Posterior credible intervals, no cluster-permutation gymnastics.
Currently running locally – included in summary if finished.
4. GAM with σ and R^2 interactions: tests whether the shift magnitude scales with PRF size (AF+ prediction) vs fit quality (artefact).
Script ready – will run after #3.

WHAT DIDN'T WORK (and why)

- Post-hoc 4-AF parametric model: $\log(g_{\text{HP}}/g_{\text{LP}})$ is bimodal at the parameter bounds (± 5), $R^2 \approx 0.005$. Underdetermined fit at this SNR. The signal is real (model-free tests confirm) but the parametric model can't recover stable parameters from per-condition PRF center estimates alone.
→ Joint AF+PRF model in braincoder (cluster plan #4) is the proper fix.
→ Methodological-paper opportunity.

OPEN QUESTIONS FOR THE MEETING

- Tunçok 2025 has a true neutral baseline (distributed attention); we don't.
Worth adding a neutral condition in future scans?
- Richter 2025 finds broad EVC suppression (HPDL + nearby neutrals). Does our PRF-shift result reflect the same proactive priority map?
- Joint AF+PRF in braincoder – methodological paper independent of retsupp?